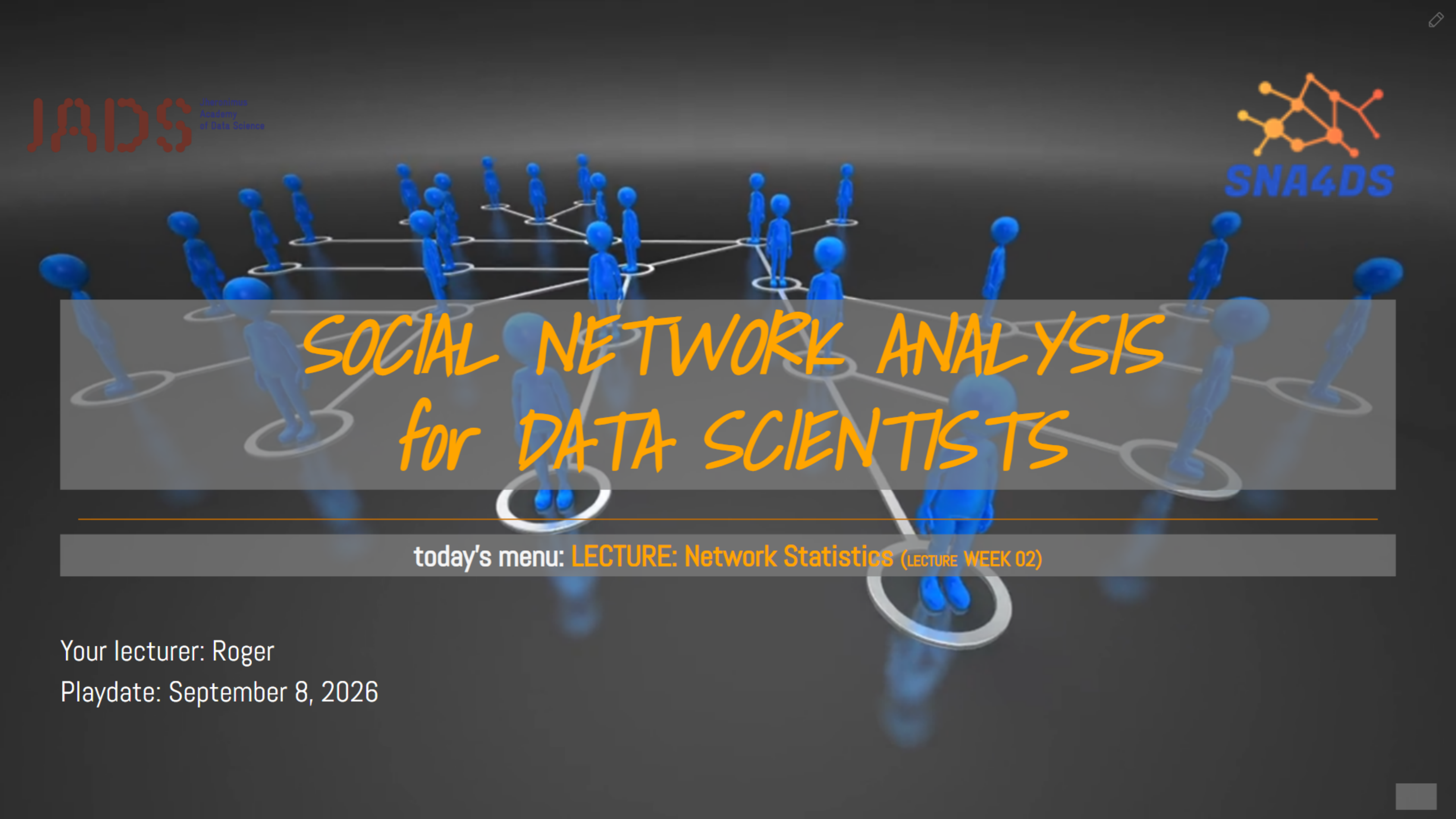




SOCIAL NETWORK ANALYSIS for DATA SCIENTISTS

today's menu: **LECTURE: Network Statistics** (LECTURE WEEK 02)

Your lecturer: Roger

Playdate: September 8, 2026

Today's menu

1. Statistical Models for Networks
2. Social Influence modeling

The course sofar



We have performed descriptive and exploratory analysis:

- vertex level indices (betweenness, closeness, pagerank, etc.)
- graph level indices (centralization, components/subgroups, et cetera)
- Getting to know R with `snafun` , `network / sna` , and `igraph`

From exploration to estimation and testing



- You found *betweenness centralization* of .2. Is this high? Low? Expected? Unexpected? Statistically significant?
- You found a 4 communities in the graph. Is this high? Low? Expected? Unexpected? Statistically significant?
- You found that more central companies innovate more. But is this expected? Unexpected? Statistically significant?
- You found a .4 correlation between friendship and advice-giving. Statistically significant?
- How to conclude if friendship promotes advice giving?

➔ we need statistics: *parameter estimation* and *hypothesis testing*.

We assume that you:



- can easily calculate the appropriate vertex-level and graph-level indices
- are comfortable with the basics of statistics (see: *bootcamp*):
 - linear regression
 - logistic regression
 - null-hypothesis testing

network autocorrelation
Conditional Uniform Graph



netlm
netlogit
ergm
gergm
btergm

bootcamp

Network statistics: Social Influence



**EVERYTHING IS RELATED TO EVERYTHING ELSE,
BUT NEAR THINGS ARE MORE RELATED
THAN DISTANT THINGS.**

Tobler's first law

The law basically says two things:

- You interact more often/strongly with whom is "near"
- Near things are more alike than distant things

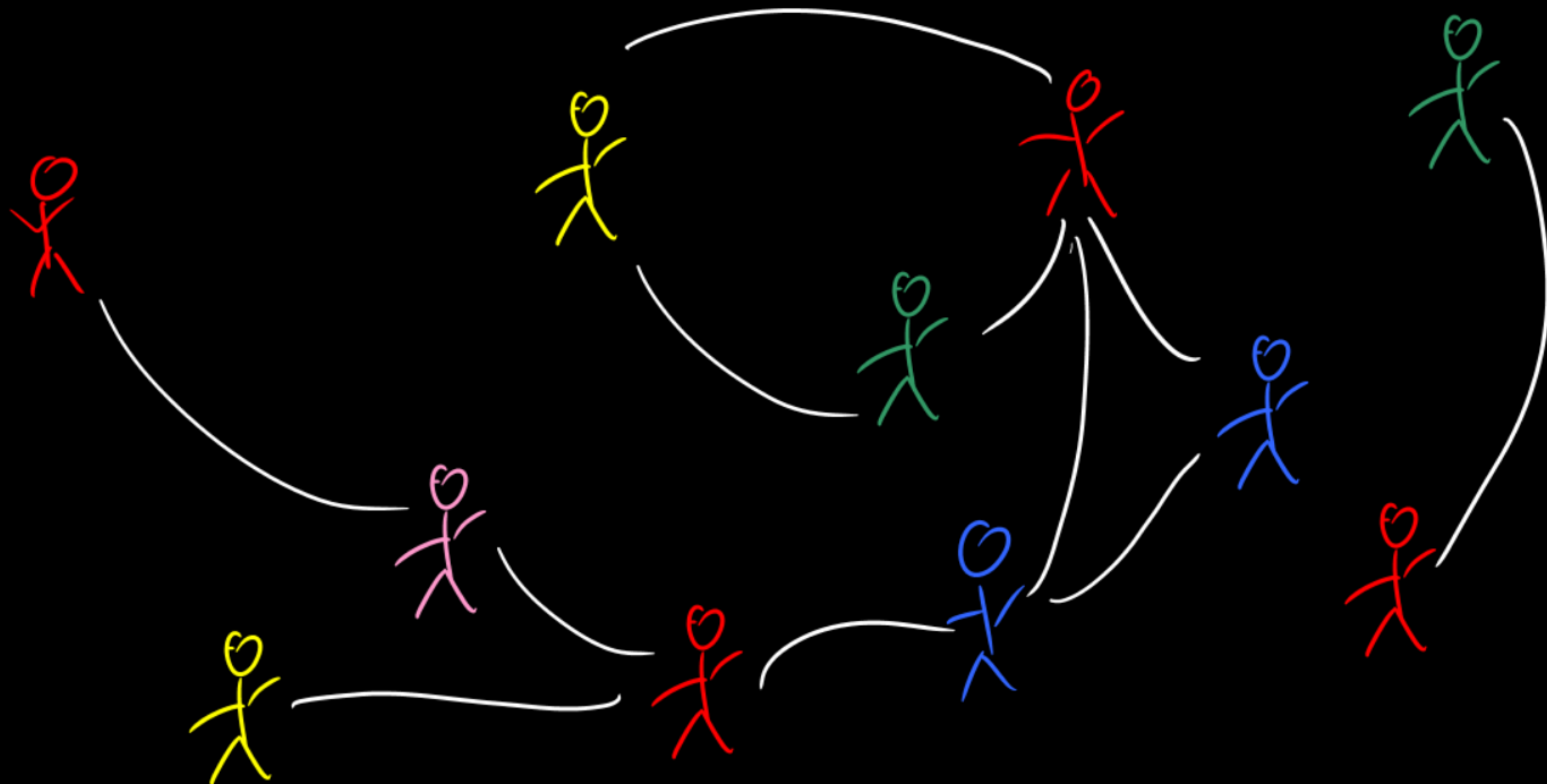


Tobler's first law of Geography

*Everything is related to everything else,
but near things are more related
than distant things.*

➔ This is the idea that social network models of social influence are built upon.

Social influence on a network



Social influence on a network

Network autocorrelation model

$$y = \rho W y + X\beta + \epsilon$$

Social influence on a network

Network autocorrelation model

vertex attribute

$$y = \rho W y + X\beta + \epsilon$$

Here, y is one's preference for a political decision, how much one likes his job, appreciation of a course, liking of a movie, propensity to buy a new model phone, the amount of CO2 one is OK with, et cetera. It is a numeric vector with a value for every actor.

Social influence on a network

Network autocorrelation model

$$y = \rho W y + X\beta + \epsilon$$

vertex attribute

intrinsic value

$X\beta$ is one's **intrinsic opinion**, the one you would hold without any social influence.

Social influence on a network

Network autocorrelation model

The diagram shows the equation $y = \rho W y + X \beta + \epsilon$ with handwritten annotations in various colors. The first y is in a blue cloud, with a blue arrow pointing to it from the text "vertex attribute". The $\rho W y$ term is in a green cloud, with a green arrow pointing to it from the text "social influence". The $X \beta$ term is in a red cloud, with a red arrow pointing to it from the text "intrinsic value".

$$y = \rho W y + X \beta + \epsilon$$

$\rho W y$ captures social influence, it moves people away from their intrinsic positions.

Social influence on a network

Network autocorrelation model

$$y = \rho W y + X \beta + \epsilon$$

This is the well-known OLS model, without any network effect (so, when: $\rho = 0$).

Social influence on a network

Network autocorrelation model

$$y = \rho W y + X \beta + \epsilon$$

$$y_1 = \rho(W_{12}y_2 + W_{13}y_3 + W_{14}y_4 + \dots + W_{1g}y_g) + X_{11}\beta_1 + X_{12}\beta_2 + \dots + X_{1k}\beta_k + \epsilon_1$$

$$y_2 = \rho(W_{21}y_1 + W_{23}y_3 + W_{24}y_4 + \dots + W_{2g}y_g) + X_{21}\beta_1 + X_{22}\beta_2 + \dots + X_{2k}\beta_k + \epsilon_2$$

$\vdots$

$$y_g = \rho(W_{g1}y_1 + W_{g3}y_3 + W_{g4}y_4 + \dots + W_{g-1,2}y_{g-1}) + X_{g1}\beta_1 + X_{g2}\beta_2 + \dots + X_{gk}\beta_k + \epsilon_g$$

g is the number of actors in the network, X a matrix of k explanatory variables, ρ a scalar comparable to a correlation coefficient.

Weight matrix W

The matrix W captures the social influence process you want to test.

Cell (i, j) captures the weight of the influence that i receives from j .

The weight matrix is usually (but not necessarily) **row-normalized** so each row adds to 1 exactly.

What does row normalization mean *substantively* here?

Mathematically, row or column normalization has the advantages of making the eigenvalues behave nicely, so the likelihood become nice and smooth.

It also restricts ρ to the $(-1, 1)$ interval, making it easy to interpret.

Tobler's first law

*Everyone is related to everyone else,
but those who are near persons are more related
than those who are distant from each other.*

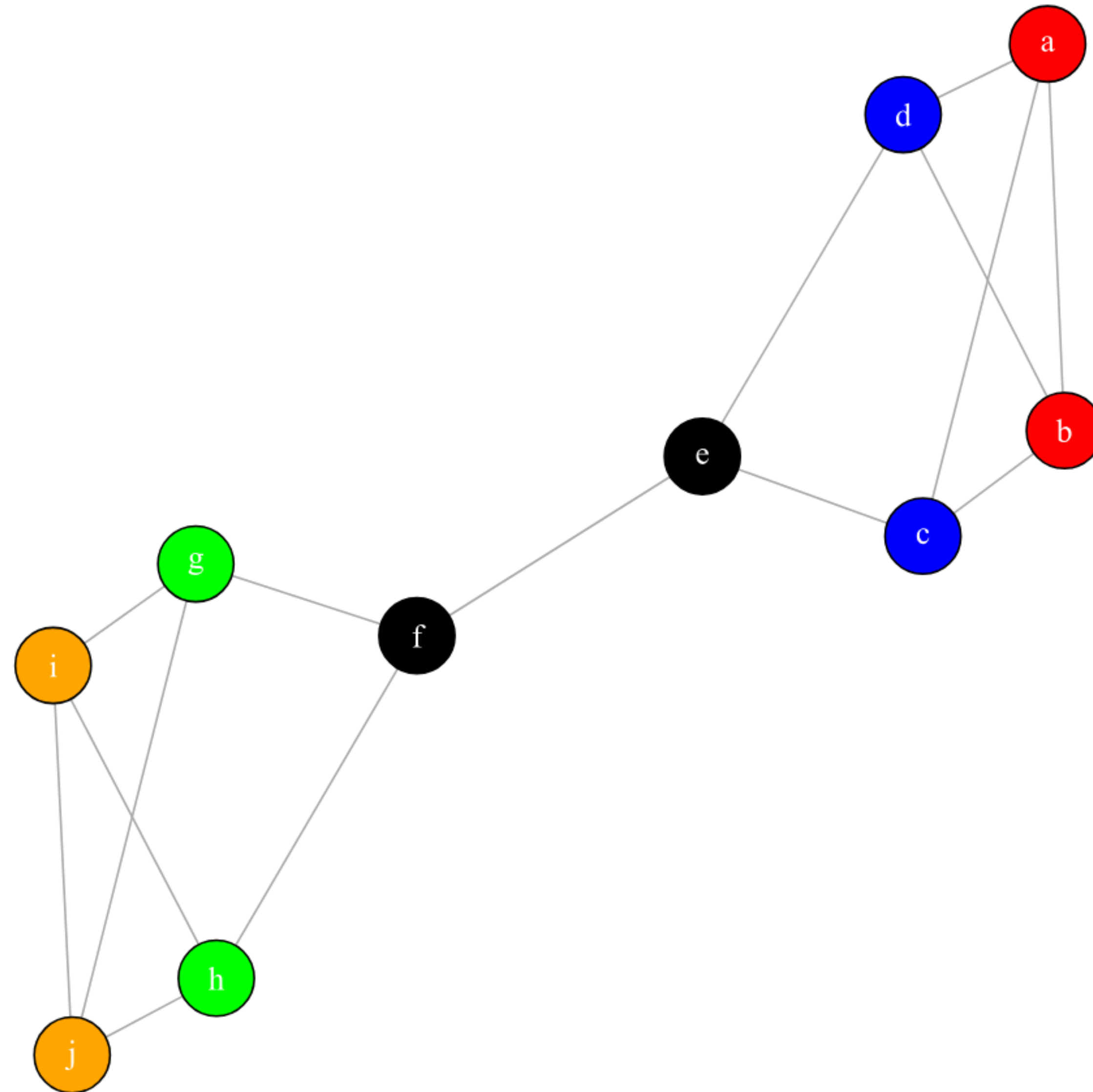
➔ What is "near" in a social network?

Two views on how people influence each other:



- communication
- comparison

Small example network



Communication



Communication refers to social influence through the **direct contact** between ego and alter.

- The more frequent and vivid the communication between ego and alter, the more likely it is that ego will adopt alter's ideas and beliefs.
- Through discussing matters with alter, ego comes to an understanding of an issue and adds new information to his own, which may cause ego to develop similar attitudes and understandings.

Comparison

- Ego compares himself to those alters whom he considers **similar** to him in relevant respects, asking himself 'what would that other person do if he were in my shoes?'
- Ego perceives (or assesses) alter's behavior and assumes that behavior to be the 'correct' behavior for 'a-person-like-me' or for 'a-person-in-a-position-like-mine'.

This is the world of "influencers" and "role models"

(whether they are aware of having this role or not)

Operationalizing COMMUNICATION

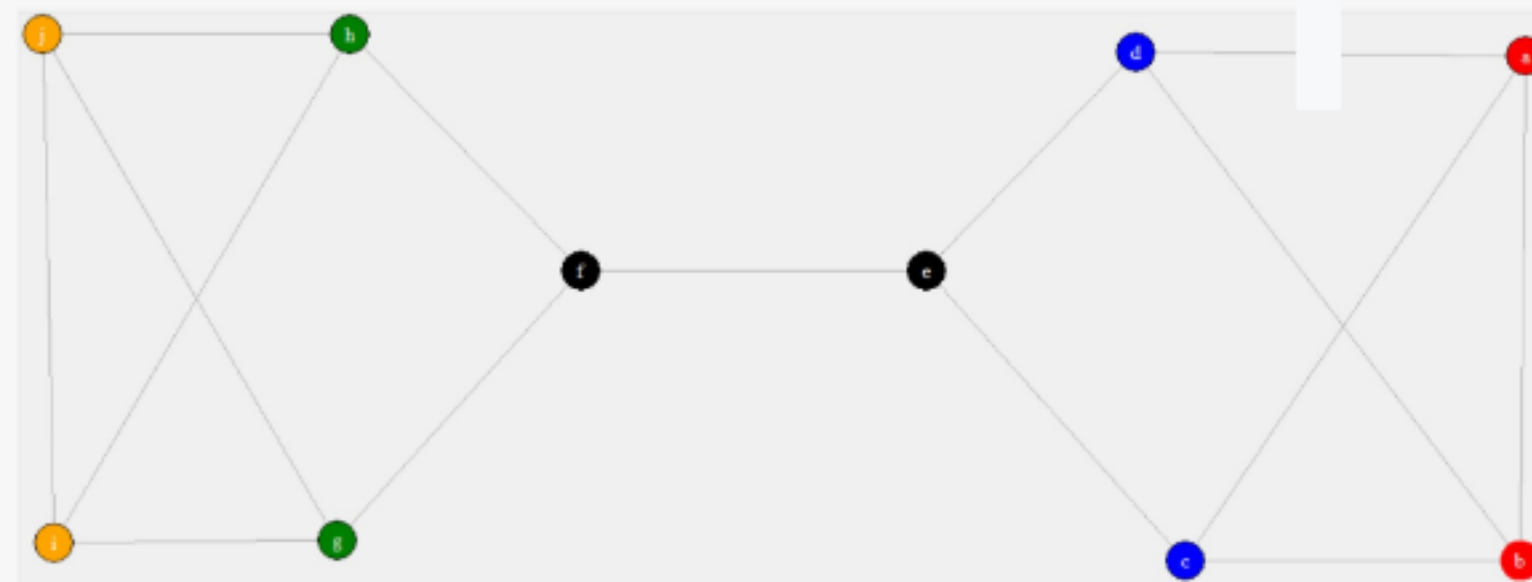
- The most straightforward weight matrix is based on the adjacency matrix: you are influenced by those you are tied to directly.

```
net_mat
```

	a	c	d	b	e	f	g	h	i	j
a	0	1	1	1	0	0	0	0	0	0
c	1	0	0	1	1	0	0	0	0	0
d	1	0	0	1	1	0	0	0	0	0
b	1	1	1	0	0	0	0	0	0	0
e	0	1	1	0	0	1	0	0	0	0
f	0	0	0	0	1	0	1	1	0	0
g	0	0	0	0	0	1	0	0	1	1
h	0	0	0	0	0	1	0	0	1	1
i	0	0	0	0	0	0	1	1	0	1
j	0	0	0	0	0	0	1	1	1	0

```
round(net_mat / rowSums(net_mat), digits = 2)
```

	a	c	d	b	e	f	g	h	i	j
a	0.00	0.33	0.33	0.33	0.00	0.00	0.00	0.00	0.00	0.00
c	0.33	0.00	0.00	0.33	0.33	0.00	0.00	0.00	0.00	0.00
d	0.33	0.00	0.00	0.33	0.33	0.00	0.00	0.00	0.00	0.00
b	0.33	0.33	0.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00
e	0.00	0.33	0.33	0.00	0.00	0.33	0.00	0.00	0.00	0.00
f	0.00	0.00	0.00	0.00	0.33	0.00	0.33	0.33	0.00	0.00
g	0.00	0.00	0.00	0.00	0.00	0.33	0.00	0.00	0.33	0.33
h	0.00	0.00	0.00	0.00	0.00	0.33	0.00	0.00	0.33	0.33
i	0.00	0.00	0.00	0.00	0.00	0.00	0.33	0.33	0.00	0.33
j	0.00	0.00	0.00	0.00	0.00	0.00	0.33	0.33	0.33	0.00



Operationalizing COMMUNICATION



There are many alternatives, such as:

- are you members of the same component/subgroup?
- number of two-paths between two people

Here, you assume that being in the same work component, advice-sharing component, collaboration component, course team, sitting at the same table, et cetera, makes it more likely that people communicate directly.

Operationalizing COMPARISON



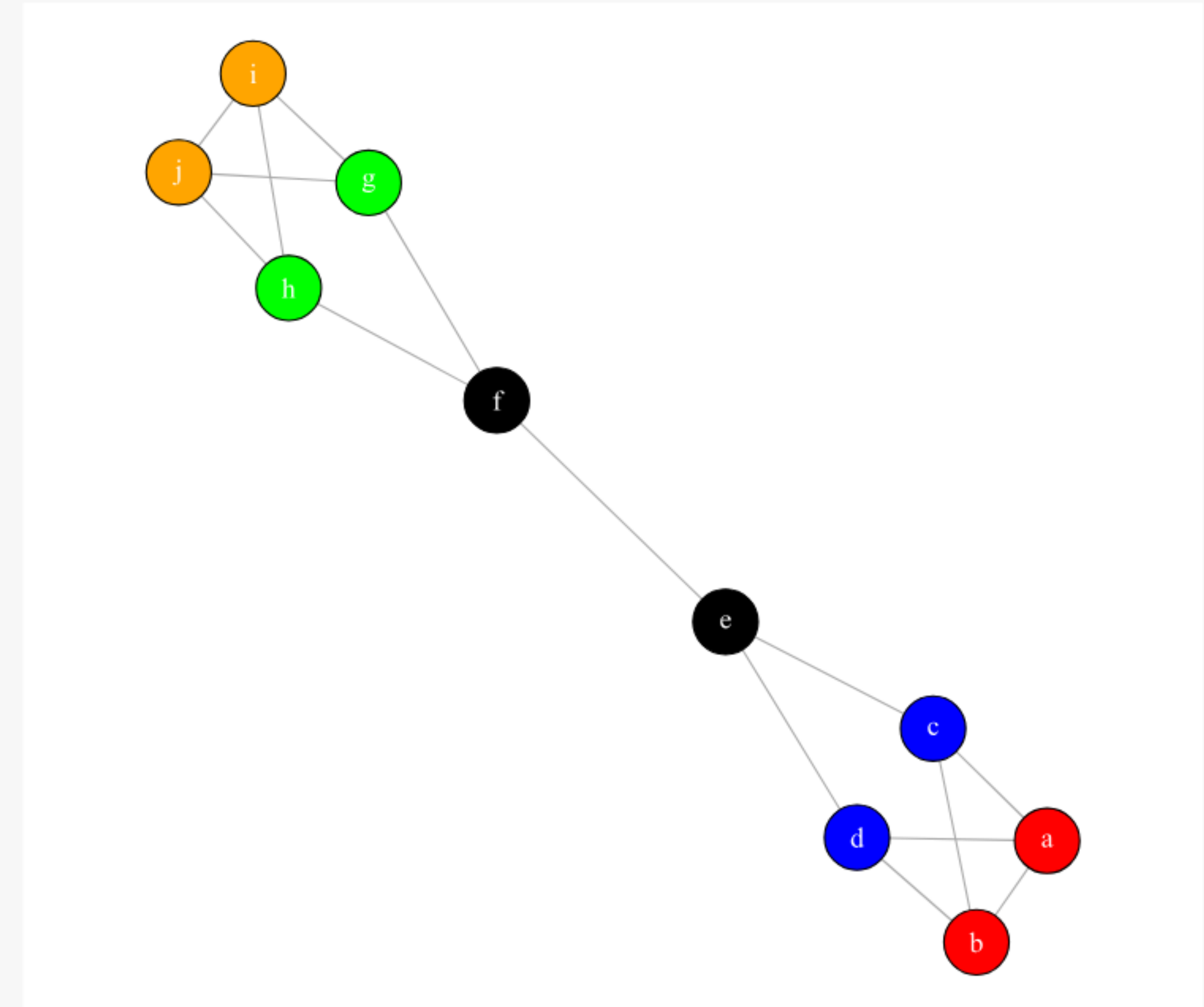
How would you operationalize whom a network actor would compare himself/herself to?

Structural equivalence

One mathematical way to measure how similar the other person is to you-- in a "social network sense", is through *structural equivalence*.

Two nodes are structurally equivalent if they have **precisely** the same relations across all other nodes in the network.

Compare a and b (in red) with e and f (in black). Which pairs are more strongly structurally equivalent?



Structural equivalence

```
distances <- snafun::d_structural_equivalence(net_sna)
distances
```

	a	c	d	b	e	f	g	h	i	j
a	1.000	0.333	0.333	1.000	0.467	-0.600	-0.600	-0.600	-0.600	-0.600
c	0.333	1.000	1.000	0.333	-0.333	-0.067	-0.600	-0.600	-0.600	-0.600
d	0.333	1.000	1.000	0.333	-0.333	-0.067	-0.600	-0.600	-0.600	-0.600
b	1.000	0.333	0.333	1.000	0.467	-0.600	-0.600	-0.600	-0.600	-0.600
e	0.467	-0.333	-0.333	0.467	1.000	-0.333	-0.067	-0.067	-0.600	-0.600
f	-0.600	-0.067	-0.067	-0.600	-0.333	1.000	-0.333	-0.333	0.467	0.467
g	-0.600	-0.600	-0.600	-0.600	-0.067	-0.333	1.000	1.000	0.333	0.333
h	-0.600	-0.600	-0.600	-0.600	-0.067	-0.333	1.000	1.000	0.333	0.333
i	-0.600	-0.600	-0.600	-0.600	-0.600	0.467	0.333	0.333	1.000	1.000
j	-0.600	-0.600	-0.600	-0.600	-0.600	0.467	0.333	0.333	1.000	1.000

This measure runs between $(-1, 1)$, with 1 showing two vertices having perfectly identical edges and -1 showing exactly opposite positions.

Running the autocorrelation model

$$y = \rho W y + X\beta + \epsilon$$

```
snafun::stat_nam(y ~ x, W = NULL, model = "lag")
```

Here, y is the dependent variable (a vector), X is the matrix with exogenous variables, and W is the weight matrix.

The model implemented in `snafun::stat_nam`

$$\begin{aligned}y &= \rho_1 W y + X\beta + \epsilon \\ \epsilon &= \rho_2 W_2 \epsilon + \nu\end{aligned}$$

What does the model represent if you set W to 0?

$$\begin{aligned}y &= X\beta + \epsilon \\ \epsilon &= \rho_2 W_2 \epsilon + \nu\end{aligned}$$

You run the models as follows:

Code	Model
<code>snafun::stat_nam(y ~ x, W = NULL, model = 'lag')</code>	$W = W, W2 = 0$
<code>snafun::stat_nam(y ~ x, W = NULL, model = 'error')</code>	$W = 0, W2 = W2$
<code>snafun::stat_nam(y ~ x, W = NULL, model = 'combined')</code>	$W = W, W2 = W2$

Running the autocorrelation model

Data

R code

Output

- 100 people read a piece of information and share their thoughts with others through a chatboard about how trustworthy they consider the information
- after the discussion, they each score the trustworthiness of the information on a continuous scale between 0 and 10
- Explanatory variables:
 - political preference
 - proportion of waking time spent on social media
 - expertise on the topic
 - agreeableness (personality trait)

Running the autocorrelation model

Data

R code

Output

```
# communication: direct ties
# trust_net is the adjacency matrix of the communication
diag(trust_net) <- 0
w_adj <- trust_net / rowSums(trust_net)

nam <- snafun::stat_nam(trust_y ~ ., W = w_adj, data = trust)
summary(nam)
```

Running the autocorrelation model

Data	R code	Output
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```
Call:(function (formula, data = list(), listw, na.action, Durbin,
  type, method = "eigen", quiet = NULL, zero.policy = NULL,
  interval = NULL, tol.solve = .Machine$double.eps, trs = NULL,
  control = list()) {  timings <- list()  .ptime_start <- proc.time()
  con <- list(tol.opt = .Machine$double.eps^0.5, fdHess = NULL,
    optimHess = FALSE, optimHessMethod = "optimHess", compiled_sse = FALSE,
    Imult = 2, cheb_q = 5, MC_p = 16L, MC_m = 30L, super = NULL,
    spamPivot = "MMD", in_coef = 0.1, type = "MC", correct = TRUE,
    trunc = TRUE, SE_method = "LU", nrho = 200, interpn = 2000,
    small_asy = TRUE, small = 1500, SEldet = NULL, LU_order = FALSE,
    pre_eig = NULL, OrdVsign = 1)  nmsC <- names(con)
  con[(namc <- names(control))] <- control
  if (length(noNms <- namc[!namc %in% nmsC]))
    warning("unknown names in control: ", paste(noNms, collapse = ", "))
  if (is.null(quiet))
    quiet <- !get("verbose", envir = .spatialregOptions)
  stopifnot(is.logical(quiet))  if (is.null(zero.policy))
    zero.policy <- get("zero.policy", envir = .spatialregOptions)
```